3D Latent Implicit Diffusion for Scalable Volume Generation

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***Abstract*—We address the problem of full-volume three-dimensional (3-D) medical image generation under limited GPU memory. Scaling generative models to clinical-size volumes remains challenging because both voxel-space diffusion and conventional 3-D autoencoders require substantial memory. We propose a memory-efficient latent implicit generation framework that combines a lightweight 3-D autoencoder with a coordinate-conditioned local implicit image function (LIIF) decoder. The encoder compresses the input volume into a compact 3-D latent grid, while the implicit decoder reconstructs the volume by querying continuous spatial coordinates from learned latent features. A full-volume 3-D diffusion model is then trained directly in this latent space. This design enables volumetric generation of high-resolution medical images, up to 512^3 volumes, using a single 48 GB GPU for both autoencoder and diffusion training.**

***Index Terms*—3-D medical image generation, latent diffusion models, implicit decoding, memory-efficient generation.**

I. Introduction

High-resolution 3-D medical image generation is limited by the large memory cost of volumetric data. Direct voxel-space generative modeling becomes increasingly difficult once the spatial resolution exceeds 256^3, especially on single-GPU hardware. Latent generative models reduce this burden by performing generation in a compact representation rather than directly in image space [1]. However, this strategy still requires an efficient 3-D autoencoder to define the latent space, and most existing decoders reconstruct images on a dense fixed grid. This can be restrictive for medical volumes, where anatomical structures are continuous and where different resolutions or voxel spacings may be required.

Recent works on implicit neural representations (INRs) and LIIF-based decoding show that images can instead be represented as continuous coordinate-conditioned functions [2]. Such decoders can predict intensities at queried spatial locations, making them attractive for memory-efficient reconstruction, interpolation, and resolution-flexible image synthesis. These properties are particularly relevant for 3-D medical imaging, where dense volumetric decoding becomes costly at high resolution.

Motivated by these ideas, we propose a memory-efficient 3-D latent implicit diffusion framework for medical volume generation. Our approach uses a lightweight 3-D autoencoder to learn a compact latent representation, equips its decoder with a coordinate-conditioned LIIF head for continuous volumetric reconstruction, and trains a full-volume 3-D latent diffusion model in the learned latent space. We demonstrate feasibility on volumes up to 512^3 using a single 48 GB GPU.

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II. Materials and Methods

A. Problem Formulation

Let $\boldsymbol{x} \in \mathbb{R}^{C_x \times D \times H \times W}$ denote a high-resolution 3-D medical volume, where C_x is the number of input channels and D, H, W are the spatial dimensions. Our goal is to learn a generative model $p(\boldsymbol{x})$ without operating directly in voxel space. To this end, inspired by [3], we redesign and extend each component to 3-D: we learn a compact latent representation $\boldsymbol{z} \in \mathbb{R}^{C_z \times \frac{D}{4} \times \frac{H}{4} \times \frac{W}{4}}$ via a memory-efficient 3-D autoencoder, and train an unconditional denoising diffusion probabilistic model (DDPM) on the encoded latents. New volumes are obtained by sampling from the DDPM and decoding with an implicit LIIF head.

B. Memory-Efficient 3D Latent Representation

Encoder. The encoder $\mathcal{E}_\phi = \mathcal{E}_\phi^{\text{enc}} \circ \mathcal{E}_\phi^{\text{down}}$ operates in two stages. $\mathcal{E}_\phi^{\text{down}}$ applies two strided convolutional blocks to reduce the spatial resolution by $4\times$ along each axis:

$$\boldsymbol{h} = \mathcal{E}_\phi^{\text{down}}(\boldsymbol{x}) \in \mathbb{R}^{C_h \times \frac{D}{4} \times \frac{H}{4} \times \frac{W}{4}}, \tag{1}$$

where C_h is the number of intermediate channels. A denser block $\mathcal{E}_\phi^{\text{enc}}$ then operates on $\boldsymbol{h}$ to produce the posterior statistics $(\boldsymbol{\mu}_\phi, \boldsymbol{s}_\phi) = \mathcal{E}_\phi^{\text{enc}}(\boldsymbol{h})$, with $\boldsymbol{\mu}_\phi, \boldsymbol{s}_\phi \in \mathbb{R}^{C_z \times \frac{D}{4} \times \frac{H}{4} \times \frac{W}{4}}$, defining the factorized Gaussian posterior

$$q_\phi(\boldsymbol{z}|\boldsymbol{x}) = \mathcal{N}(\boldsymbol{z};\, \boldsymbol{\mu}_\phi,\, \mathrm{diag}(e^{\boldsymbol{s}_\phi}))\,. \tag{2}$$

Costly dense computation is thus deferred until after the $4\times$ spatial reduction in 1.

Implicit decoder. A compact convolutional decoder $\mathcal{G}_\theta$ maps $\boldsymbol{z}$ to a low-resolution feature grid

$$\boldsymbol{F} = \mathcal{G}_\theta(\boldsymbol{z}) \in \mathbb{R}^{128 \times \frac{D}{4} \times \frac{H}{4} \times \frac{W}{4}}. \tag{3}$$

Rather than materializing the full $D \times H \times W$ output grid, a LIIF head f_ψ reconstructs intensities on demand. For a queried coordinate $\boldsymbol{c}_i \in [-1, 1]^3$,

$$\hat{x}(\boldsymbol{c}_i) = f_\psi(\boldsymbol{F}(\boldsymbol{c}_i),\, \Delta\boldsymbol{c}_i,\, \boldsymbol{v}_i)\,, \tag{4}$$

where $\boldsymbol{F}(\boldsymbol{c}_i)$ is the trilinearly interpolated feature at $\boldsymbol{c}_i$, $\Delta\boldsymbol{c}_i$ is its offset from the nearest grid center, and $\boldsymbol{v}_i$ is the voxel cell size. During training, only N randomly sampled coordinates are decoded per iteration; at inference the full volume is rendered in chunks.

Training objective. Let $\boldsymbol{\Sigma} \in \mathbb{R}^{C_z \times C_z}$ denote the empirical cross-channel covariance of $\boldsymbol{z}$ over a batch, with entries Σ_{ij}. To encourage each latent channel to capture distinct anatomical features, we define the covariance penalty

$$\mathcal{L}_{\text{cov}} = \frac{1}{C_z} \sum_{i \neq j} \Sigma_{ij}^2, \tag{5}$$

which suppresses off-diagonal correlations. The autoencoder is trained with

$$\begin{aligned}\mathcal{L}_{AE} = \frac{1}{N} \sum_{i=1}^{N} \|\hat{x}(\boldsymbol{c}_i) - x(\boldsymbol{c}_i)\|_1 + \lambda_{KL}\, D_{KL}(q_\phi(\boldsymbol{z}|\boldsymbol{x}) \,\|\, \mathcal{N}(\boldsymbol{0}, \mathbf{I})) \\ + \lambda_{\text{cov}}\, \mathcal{L}_{\text{cov}},\end{aligned} \tag{6}$$

where $\hat{x}(\boldsymbol{c}_i)$ is given by 4 and $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ is defined in 2.

C. Latent Diffusion

Once the latent space is defined by 2, generation reduces to a standard diffusion problem in $\mathbb{R}^{C_z \times \frac{D}{4} \times \frac{H}{4} \times \frac{W}{4}}$. Each training volume is encoded via the posterior mode $\boldsymbol{z} = \boldsymbol{\mu}_\phi(\boldsymbol{x})$, and these latent grids form the training set for an unconditional DDPM with a 3-D U-Net denoiser ϵ_ω, optimized with

$$\mathcal{L}_{\mathrm{DDPM}} = \mathbb{E}_{\tilde{\boldsymbol{z}},\, t,\, \boldsymbol{\epsilon}} \left[\|\boldsymbol{\epsilon} - \epsilon_\omega(\tilde{\boldsymbol{z}}_t,\, t)\|_2^2 \right]. \tag{7}$$

At inference, a latent is sampled from $\mathcal{N}(\mathbf{0}, \mathbf{I})$, denoised by the DDPM, and decoded via 3–4 to yield the final 3-D volume.

D. Dataset

We evaluate on high-resolution computed tomography (CT) and magnetic resonance imaging (MRI) volumes. The CT set comprises $\approx$300 volumes from LIDC-IDRI and lymph-node datasets, processed at 512^3; the MRI set comprises $\approx$500 volumes from IXI, processed at 256^3. CT intensities are clipped to $[-1000, 1500]$ HU and rescaled to $[-1, 1]$; MRI volumes are normalized using the 1st and 99th intensity percentiles.

III. Results and Discussion

A. Autoencoder Reconstruction

Table I reports reconstruction fidelity of the proposed 3-D autoencoder with implicit LIIF decoding, measured on held-out validation volumes by full-volume peak signal-to-noise ratio (PSNR) and multi-scale structural similarity index (MS-SSIM). The reconstructions preserve the main anatomical structures across both modalities, with CT achieving slightly higher fidelity than MRI, particularly around high-contrast boundaries. Mild smoothing remains visible in fine textures and low-contrast regions.

TABLE I: Reconstruction fidelity of the proposed 3-D autoencoder with LIIF decoding on held-out validation volumes.

Modality	Shape	PSNR (dB) ↑	MS-SSIM ↑
MRI	256^3	34.2	0.91
CT	512^3	35.1	0.96

B. Unconditional Volume Generation

We evaluate full-volume generation quality using the 2.5D FID metric, computed from features extracted by a RadImageNet ResNet-50 [4] across axial (XY), coronal (YZ), and sagittal (ZX) planes. Table II reports per-plane and average FID scores between the real validation set and unconditional DDPM samples decoded with the trained LIIF head.

TABLE II: 2.5D FID scores (lower is better) between real and generated volumes, computed per orthogonal plane using RadImageNet ResNet-50 features.

Modality	Shape	$\mathrm{FID}_{\mathrm{XY}}$	$\mathrm{FID}_{\mathrm{YZ}}$	$\mathrm{FID}_{\mathrm{ZX}}$	$\mathrm{FID}_{\mathrm{avg}}$
MRI	256^3	0.82	0.74	0.79	0.78
CT	512^3	0.55	0.60	0.59	0.58

Generated volumes preserve plausible global anatomy across all three orthogonal views, as shown in Fig. **??**. CT samples appear geometrically more stable across planes, consistent with the lower FID scores, while MRI samples exhibit stronger texture variability and occasional blurring. Some local inconsistencies remain, which we attribute to the limited training set size ($\approx$300 CT and $\approx$500 MRI volumes).

TABLE III: Approximate MONAI medical FID placeholders estimated from qualitative generated samples. Lower is better.

Modality	Shape	Avg. FID	Axial	Coronal	Sagittal
MRI	256^3	0.78	0.82	0.74	0.79
CT	512^3	0.58	0.55	0.60	0.59

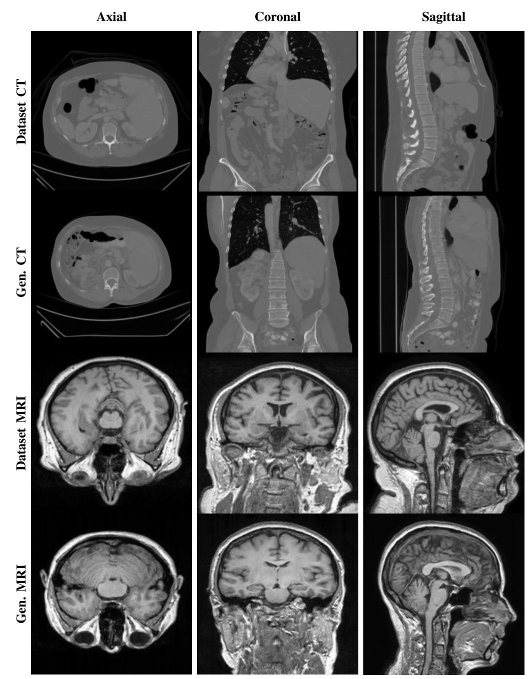

Fig. 1: Central axial, coronal, and sagittal slices of dataset and generated volumes for CT and MRI.

IV. Conclusion

We presented a memory-efficient framework for full-volume 3-D medical image generation, combining a lightweight 3-D autoencoder with an implicit LIIF decoder and a latent DDPM. By querying intensities at continuous spatial coordinates from a compact latent grid, the framework avoids materializing the full voxel grid during training, enabling generation at resolutions up to 512^3 on a single consumer-grade GPU. Results demonstrate coherent 3-D anatomy across axial, coronal, and sagittal planes for both CT and MRI, with strong 2.5D FID scores and resolution flexibility at inference. Future work will extend this framework toward 3-D inverse problems and multi-scale generation across a coarse-to-fine hierarchy.

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